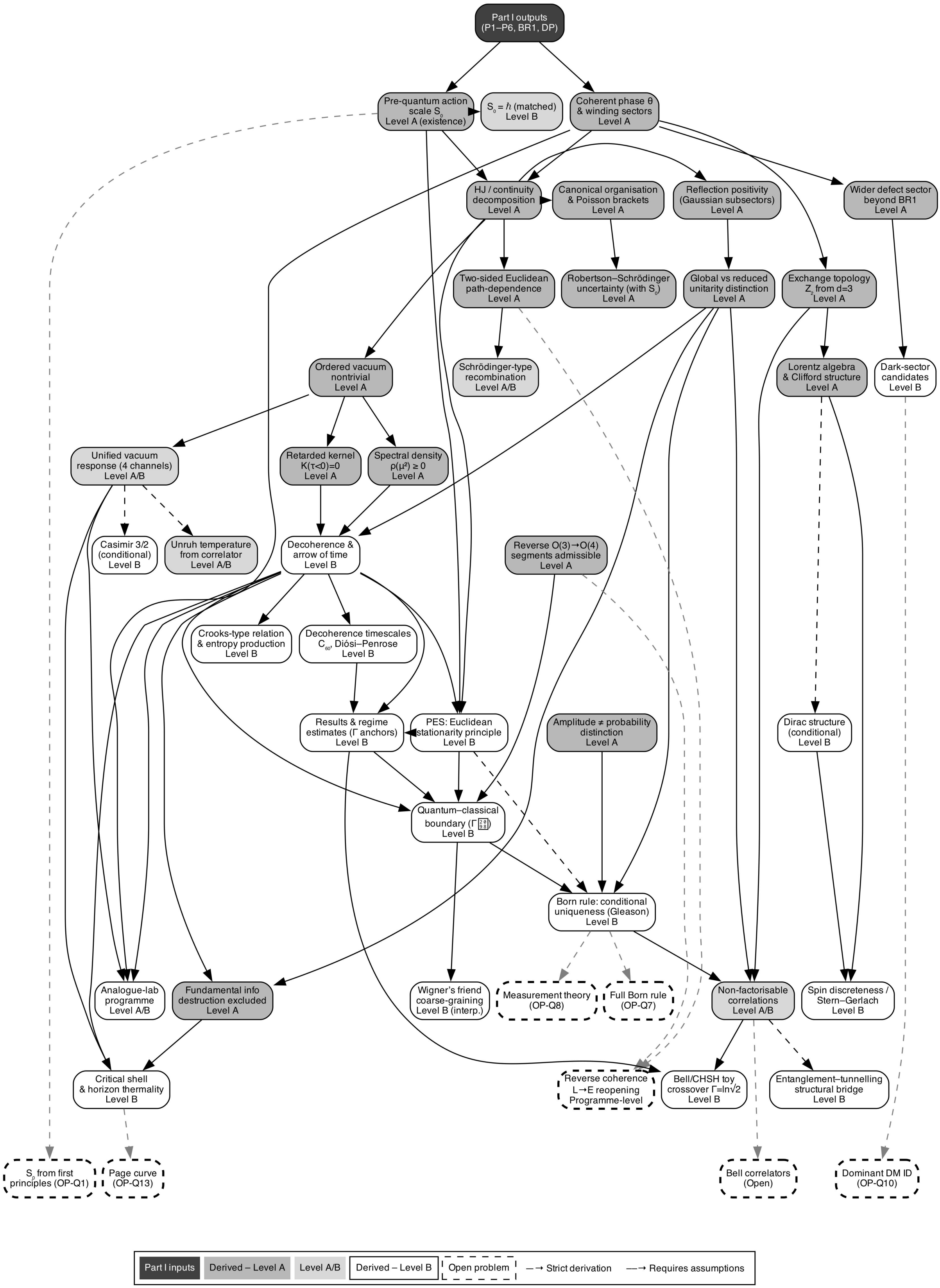
Part I outputs (P1–P6, BR1, DP)
Pre-quantum action scale S_0 Level A (existence)
$S_0 = \hbar$ (matched) Level B
Coherent phase θ & winding sectors Level A
HJ / continuity decomposition Level A
Canonical organisation & Poisson brackets Level A
Reflection positivity (Gaussian subsectors) Level A
Wider defect sector beyond BR1 Level A
Two-sided Euclidean path-dependence Level A
Robertson–Schrödinger uncertainty (with S_0) Level A
Global vs reduced unitarity distinction Level A
Exchange topology Z_2 from d=3 Level A
Ordered vacuum nontrivial Level A
Schrödinger-type recombination Level A/B
Lorentz algebra & Clifford structure Level A
Dark-sector candidates Level B
Unified vacuum response (4 channels) Level A/B
Retarded kernel K(τ<0)=0 Level A
Spectral density ρ(μ²) ≥ 0 Level A
Casimir 3/2 (conditional) Level B
Unruh temperature from correlator Level A/B
Decoherence & arrow of time Level B
Reverse O(3)→O(4) segments admissible Level A
Crooks-type relation & entropy production Level B
Decoherence timescales C_{60}, Diósi–Penrose Level B
Results & regime estimates (Γ anchors) Level B
PES: Euclidean stationarity principle Level B
Amplitude ≠ probability distinction Level A
Dirac structure (conditional) Level B
Quantum–classical boundary (Γ 20/98) Level B
Born rule: conditional uniqueness (Gleason) Level B
Analogue-lab programme Level A/B
Fundamental info destruction excluded Level A
Wigner's friend coarse-graining Level B (interp.)
Measurement theory (OP-Q8)
Full Born rule (OP-Q7)
Non-factorisable correlations Level A/B
Spin discreteness / Stern–Gerlach Level B
Critical shell & horizon thermality Level B
Reverse coherence L→E reopening Programme-level
Bell/CHSH toy crossover Γ=ln√2 Level B
Entanglement–tunnelling structural bridge Level B
S_0 from first principles (OP-Q1)
Page curve (OP-Q13)
Bell correlators (Open)
Dominant DM ID (OP-Q10)
Part I inputs
Derived – Level A
Level A/B
Derived – Level B
Open problem
Strict derivation
Requires assumptions